Physics 7A: Mechanics

Fall 2026 Problem Set 4: Forces, Energy and Calculus Review

Due Friday, November 13

1. Block on an incline

A block of mass $m = 2.0\,\text{kg}$ rests on a frictionless ramp inclined at angle $\theta = 30°$, as shown in Figure 1.

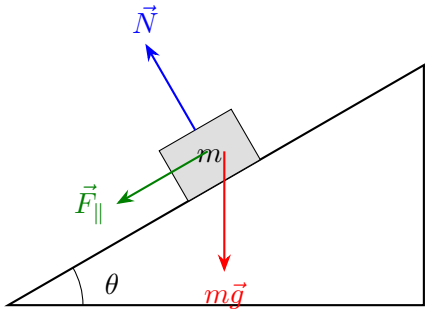


Figure 1: Free-body diagram of a block on a frictionless incline.

(a) Show that the acceleration down the ramp is

$$a = g \sin\theta. \tag{1}$$

(b) Using energy conservation, show that the speed after sliding a distance d is

$$v = \sqrt{2gd\sin\theta}. \tag{2}$$

2. Calculus review

(a) Evaluate the integral

$$\int_0^{\pi} x^2 \sin x \, dx.$$

(b) Find the limit

$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n.$$

(c) A particle's position is $x(t) = Ae^{-\gamma t}\cos(\omega t + \phi)$. Find $\dfrac{dx}{dt}$.

3. Measured data

A cart is released from rest on a track. Table 1 lists its measured position.

Table 1: Position of the cart over time.

Time t (s)	Position x (m)	Uncertainty σ_x (m)
0.0	0.00	0.01
0.5	0.31	0.01
1.0	1.22	0.02
1.5	2.76	0.02
2.0	4.90	0.03

(a) Fit the data to $x(t) = \tfrac{1}{2}at^2$ by minimizing

$$\chi^2 = \sum_{i=1}^{N} \frac{\left(x_i - \tfrac{1}{2}at_i^2\right)^2}{\sigma_i^2}.$$

(b) Show that the best-fit acceleration is

$$\hat{a} = \frac{2\sum_i t_i^2 x_i/\sigma_i^2}{\sum_i t_i^4/\sigma_i^2}.$$

4. Matrices and oscillators

Two coupled masses obey $\ddot{\mathbf{x}} = -M\mathbf{x}$, where

$$M = \frac{k}{m}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

(a) Find the eigenvalues $\lambda_{1,2}$ of M and the normal-mode frequencies $\omega_{1,2} = \sqrt{\lambda_{1,2}}$.

(b) The total energy is $E = \tfrac{1}{2}m(\dot{x}_1^2 + \dot{x}_2^2) + \tfrac{1}{2}k\left[x_1^2 + (x_2 - x_1)^2 + x_2^2\right]$. Show that $\dfrac{dE}{dt} = 0$.